

Student Information

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Table of Contents

1. Task 1 – AI Prompting in 2026
 2. Task 2 – AI Generated Snake / Worms Zone Game Development Journey
 3. Task 3 – Lecture Slide Generation Using LLMs
 4. Overall Learning Outcomes
 5. Final Conclusion
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Task 1 – AI Prompting in 2026

Objective

The objective of this lab was to understand modern AI prompting techniques and how different prompt structures affect AI-generated outputs.

Summary of Learning

This lab demonstrated that prompting is not just asking questions, but designing structured instructions for AI systems.

Key concepts learned:

- Clear instructions improve output quality
- Context improves relevance
- Role-based prompting enhances specialization
- Structured outputs improve usability
- Iterative refinement improves results

Challenges Faced

- Receiving generic responses from vague prompts
- Inconsistent formatting in outputs
- Overly long or irrelevant responses
- Difficulty in refining prompts initially

Key Takeaway

Better prompts lead to better AI results. Prompt engineering is an essential skill for working effectively with AI systems.

Task 2 – AI Generated Snake / Worms Zone Game Development

Objective

To design, develop, debug, and deploy a browser-based Snake / Worms Zone-style game using AI tools.

Tools Used

- ChatGPT
- Google AI Studio
- GitHub Pages
- GitHub Actions

Development Journey

Step 1 – Basic Snake Game (ChatGPT)

- Created simple grid-based Snake game
 - Basic movement system
 - Simple scoring system
 - Minimal UI design
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Step 2 – Intermediate Version (Google AI Studio)

- Improved gameplay logic
- Better UI design
- Successfully deployed using GitHub Pages



Live Project:

<https://umairsohail92.github.io/AI-101-TASK2-SNAKE-GAME/>

Step 3 – Worms Zone Inspired Game

Developed an advanced version inspired by **Worms Zone .io** game mechanics.



Final Game Link:

<https://umairsohail92.github.io/A1-101-task-1-snake-advancegame/>

Step 4 – Debugging Phase

Issues faced:

- Snake movement errors
- Cursor control problems
- Game instability

These were fixed using iterative AI-assisted debugging.

Step 5 – Deployment

Final deployment achieved using:

- GitHub Pages

- GitHub Actions
- Correct repository configuration

✓ Final game successfully published online

Challenges Faced

- Understanding AI-generated code structure
 - Fixing JavaScript logic errors
 - Debugging movement and controls
 - Learning GitHub deployment process
 - Managing iterative improvements
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Key Learnings

- AI-assisted development workflow
 - JavaScript game logic fundamentals
 - Debugging AI-generated code
 - GitHub deployment and hosting
 - Iterative development process
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Final Outcome

- Successfully built multiple game versions
 - Upgraded Snake game into Worms Zone-style game
 - Learned debugging and deployment workflows
 - Deployed a fully working browser game
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Task 3 – Lecture Slide Generation Using LLMs

Objective

To use a Large Language Model (NotebookLM) to analyze a lecture and generate presentation slides automatically.

Tools Used

- NotebookLM (Google)
 - AI-101 Orientation Lecture (YouTube)
 - Slide Deck Generation Feature
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Process Followed

Step 1 – Lecture Input

The AI-101 orientation lecture was added into NotebookLM.

Step 2 – Content Analysis

NotebookLM analyzed the lecture transcript and extracted key concepts.

Step 3 – Slide Generation

The built-in Slide Deck feature automatically generated a structured presentation.

Step 4 – Review Process

Slides were reviewed to ensure:

- Correct coverage of topics
 - Logical structure
 - Relevance to lecture content
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Prompts Used

No custom prompts were used.

NotebookLM's automatic Slide Deck feature handled the entire generation process.

Challenges Faced

- Ensuring correct lecture import
 - Verifying accuracy of generated slides
 - Organizing final output for submission
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Key Learnings

- LLMs can automatically convert lectures into presentations
 - AI reduces manual effort in documentation
 - Human review is still important for accuracy
 - NotebookLM is effective for academic content transformation
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Final Outcome

- Successfully generated lecture slides using LLM
 - No manual prompting required
 - Learned AI-based content transformation workflow
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Overall Learning Outcomes

Through all three tasks, I learned:

AI Skills

- Prompt engineering fundamentals
- AI-assisted development workflow
- Content generation using LLMs

Technical Skills

- JavaScript game development
- Debugging AI-generated code

- GitHub Pages deployment



Productivity Skills

- Automating presentation creation
- Converting lectures into structured slides
- Using AI for rapid prototyping



Final Conclusion

This complete lab experience demonstrated that AI is not just a tool for generating content but a powerful development assistant.

Across all tasks, I learned how to:

- Work with AI tools effectively
- Debug and improve AI-generated outputs
- Deploy real-world applications
- Transform educational content into structured outputs

Overall, this course significantly improved my understanding of AI-driven development and strengthened my practical skills in using modern AI systems for real-world problem solving.